Work-Energy Theorem & Systems

Printable student worksheet pack

Use this pack alongside the Lesson 5 interactive. Attempt each question first; use the HTML Hint button only when needed, then reveal the Worked solution.

Take g = 10 m s^{-2} throughout unless otherwise stated.

Core equations and conventions

For a single object, the work-energy theorem is:

$$\Delta K = \Sigma W_i$$

For the simplified mechanical-energy ledger used in this worksheet:

$$\Delta K + \Delta U_g + \Delta U_s = W_{ext}$$

and

$$\Delta E_{mech} = \Delta K + \Delta U_g + \Delta U_s$$

Near Earth: $\Delta U_g = mg\Delta y$. Ideal spring: $U_s = \frac{1}{2}k(\Delta x)^2$.

System rule: do not double-count an interaction. If Earth is inside the system, use ΔU_g rather than external gravitational work. If an ideal spring is inside, use ΔU_s rather than external spring work.

Friction: in the 05a worksheet, friction is generally treated as external in the mechanical-energy ledger. For sliding contact, internal-energy generation depends on relative sliding distance: $\Delta U_{int} = f_k d_{rel} \geq 0$.

Before every calculation

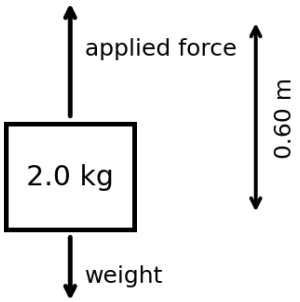
1. What physical objects are inside the system? 2. Which energy stores are relevant? 3. Which external forces do work?

Part A · 05a core worksheet

Complete the energy ledger before doing detailed calculations. Show full working underneath.

1. Lifting a 2.0 kg block

A 2.0 kg block is lifted from rest by a vertical force of 25 N through 0.60 m. Compare System A: block + Earth with System B: block only. For each, determine ΔK, ΔU_g, ΔU_s, W_{ext} and ΔE_{mech}. Hence determine the final speed.



System A: block + Earth

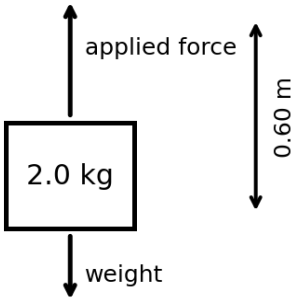
System A		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

System B: block only

System B		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

2. Lowering the block

The same 2.0 kg block is lowered from rest through 0.60 m. The person exerts a constant 15 N upward force while the displacement is downward. Repeat the analysis for block + Earth and block only, and determine the final speed.



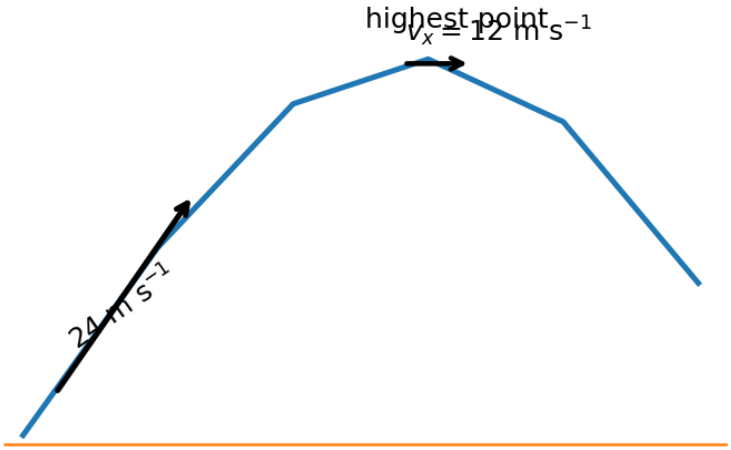
System A: block + Earth

System A		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

System B: block only

System B		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

3. Projectile and system choice



A stone is thrown from ground level at 24 m s^{-1}, 60° above the horizontal. Neglect air resistance.

(a) Determine the horizontal component of the initial velocity.

(b) Taking the stone + Earth as the system, determine the relevant terms from release to the highest point.

Stone + Earth		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

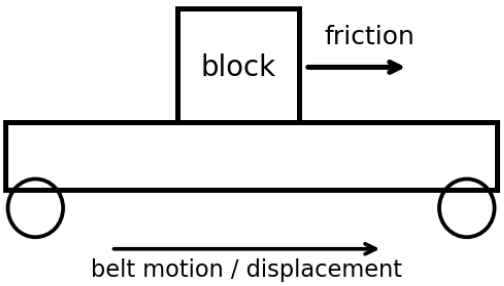
(c) Hence determine the maximum height reached.

(d) Now take the stone only as the system. Determine the relevant terms and hence determine the same maximum height.

Stone only		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}



4. Work done by friction



(a) Sliding to a stop

A 5.0 kg block is moving at 2.0 m s^{-1} along a conveyor belt. The belt is then stationary while the block slides. Kinetic friction is 16 N opposite the motion.

(i) Taking the block as the system, determine the relevant energy terms and the stopping distance.

Block		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

(ii) Determine the internal energy generated by the sliding contact.

(iii) State the mechanical work done by friction on the stationary belt and comment briefly.

4(b). Accelerating without sliding

A 5.0 kg block starts from rest on a conveyor belt. The belt accelerates it without slipping using a constant static-friction force of 20 N to the right over 0.50 m.

(i) Determine the work done on the block by friction.

(ii) Hence determine the kinetic energy gained and the final speed.

Block		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

(iii) Determine the internal energy generated by friction at the contact.

(iv) Determine the work done on the belt by friction from the block.

4(c). Accelerating with sliding

A stationary 5.0 kg block is placed on a moving conveyor belt. Kinetic friction on the block is 0.80 N to the right. The block moves 0.10 m relative to the ground before matching the constant belt speed.

(i) Determine the work done on the block by friction.

(ii) Hence determine the kinetic energy gained and the final speed.

Block		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

(iii) Determine the relative sliding distance and hence the internal energy generated.

(iv) Determine the work done on the belt by friction from the block.

(v) Draw an energy-flow diagram showing the energy supplied by the motor/belt, the block's kinetic-energy gain, and the internal-energy increase.

Part B · Quick checks

These are short conceptual checks corresponding to the interactive lesson.

1. A block falls while the system is block + Earth. Should gravity appear as external work or as ΔU_g? Explain.

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2. A block is pushed by a spring while the system is block + spring. Should you include both W_{spring} and ΔU_s? Explain.

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3. Can kinetic friction do positive work on an object? Give a physical example.

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4. Why is work done by friction on one object not generally equal to the internal energy generated at a sliding contact?

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5. A system has $\Delta K = +6$ J, $\Delta U_g = -10$ J and $\Delta U_s = +1$ J. Determine ΔE_{mech}.

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6. State the AP work-energy theorem for a single object.

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Part C · Extra Practice — 08a

For each situation, fill in the change-based energy ledger. Use the stated system boundary. Unless otherwise stated, neglect air resistance and take g = 10 m s^{-2}.

A.

A man lifts a 5 kg rucksack from the floor to a hook 2.0 m above the ground. The rucksack starts and ends at zero speed. System = rucksack only.

Situation A		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

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B.

Same physical situation as A, but system = rucksack + Earth.

Situation B		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

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C.

A girl drags a 10 kg sandbag horizontally 2.5 m with a constant horizontal pulling force of 8 N. Friction is negligible. The sandbag starts from rest. System = sandbag.

Situation C		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

D.

The girl drags the 10 kg sandbag horizontally 2.5 m with a constant 8 N pull, maintaining a constant speed of 1.0 m s^{-1}. System = sandbag.

Situation D		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

E.

The girl drags the 10 kg sandbag horizontally 2.5 m with a constant 8 N pull. Average friction is 2 N. The sandbag starts from rest. System = sandbag.

Situation E		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

F.

The girl drags the 10 kg sandbag 2.5 m up an incline, increasing its height by 0.10 m. The pulling force is a constant horizontal 8 N and average friction is 2 N. Initial speed = 1.0 m s^{-1}. System = sandbag + Earth.

Situation F		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

G.

Same physical situation as F, but system = sandbag only.

Situation G		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

H.

A 65 kg box is pushed 2.0 m up a frictionless 30° slope by a 600 N force parallel to the slope. System = box.

Situation H		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

I.

Same physical situation as H, but system = box + Earth.

Situation I		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

J.

A 65 kg box is pushed 2.0 m up a 30° slope by a 600 N force parallel to the slope. Average friction = 5 N. System = box.

Situation J		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

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K.

Same physical situation as J, but system = box + Earth. Treat friction as external in this worksheet model.

Situation K		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

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L.

A spring has k = 2.5 N cm^{-1} and relaxed length 8.0 cm. It is compressed to half its relaxed length and fires a 0.10 kg ball. System = ball.

Situation L		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

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M.

Same spring-launch situation as L, but system = ball + spring.

Situation M		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

N.

A 45 g pinball is moving up a slope at 0.64 m s^{-1} when a spring-mounted lever releases. Later it has risen 8.5 cm, is moving at 1.2 m s^{-1}, has travelled 55 cm, and experienced average friction 0.14 N. System = Earth + ball + spring.

Situation N		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

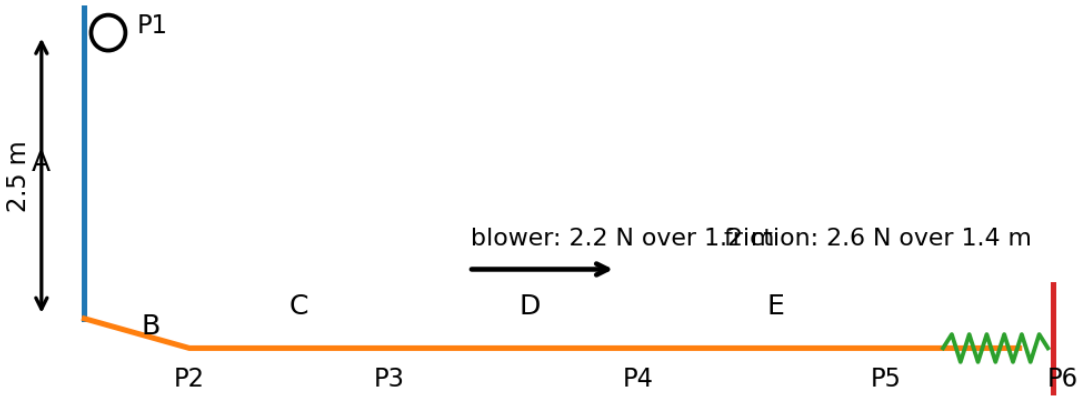
O.

Same physical situation as N, but system = ball only.

Situation O		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

Part D · Extra Practice — multi-stage track problem

The system consists of the spring + ball + Earth. Unless otherwise stated, neglect air resistance and take g = 10 m s^{-2}.



A 120 g ball is released from P1. It falls vertically 2.5 m before travelling along the horizontal track. Along stage D, a blower exerts a constant 2.2 N force to the right for 1.2 m. Along stage E, a constant friction force of 2.6 N acts for 1.4 m. The ball then compresses the spring and is momentarily at rest at P6.

1. Identify and discuss the energy transfers from P1 → P2 → P3 → P4 → P5 → P6.

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2. Write the energy equation comparing the system at P1 and P6. Determine each term, including the work done on the system.

P1 → P6		
ΔK	ΔU_g	ΔU_s
ΔE_{mech}		W_{ext}

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3. Assuming no transfer to internal energy during the subsequent decompression of the spring, calculate the speed at which the ball is projected.

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4. The spring is compressed by 2.5 cm at P6. Determine its stiffness constant k.

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5. Considering only the ball + Earth as the system, state whether the work done on the ball by each force is positive, negative or zero:

(a) gravity during stage A
(b) the slope / normal force during stage B
(c) the blower during stage D
(d) friction from the surface during stage E

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End-of-pack self-check

☐ I define the system before writing the energy equation.
☐ I do not double-count gravity or spring interactions.
☐ I can distinguish work on one object from sliding-friction dissipation.
☐ I can use ΔK, ΔU_g, ΔU_s, W_{ext} and ΔE_{mech} consistently.
☐ I state the equation before substituting numerical values.